

SYMPLECTIC SURFACES

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ABSTRACT. IAS/PCMI lecture notes.

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¹Based on PCMI lectures by Laura Starkston [24].

Part 1. Symplectic Surfaces²

1. SYMPLECTIC SURFACES AND PSEUDOHOLOMORPHIC CURVES

Throughout, (M, ω) is a symplectic four-manifold, (Σ, j) is a closed Riemann surface, and $H = [\mathbb{CP}^1] \in H_2(\mathbb{CP}^2; \mathbb{Z})$, with the normalizations below [24].

$$\begin{aligned} z_k &= x_k + iy_k, & i\partial_{x_k} &= \partial_{y_k}, & i\partial_{y_k} &= -\partial_{x_k}, & \omega_{\text{std}} &= \sum_{k=1}^n dx_k \wedge dy_k = \frac{i}{2} \sum_{k=1}^n dz_k \wedge d\bar{z}_k. \\ g_{\text{std}}(\mathbf{u}, \mathbf{v}) &= \omega_{\text{std}}(\mathbf{u}, i\mathbf{v}), & S^5 &= \{\mathbf{z} \in \mathbb{C}^3 \mid |\mathbf{z}| = 1\}, & \iota: S^5 &\hookrightarrow \mathbb{C}^3, & \pi: S^5 &\rightarrow S^5/S^1 = \mathbb{CP}^2. \\ \pi^* \omega_{FS} &= \iota^* \omega_{\text{std}}, & H_2(\mathbb{CP}^2; \mathbb{Z}) &= \mathbb{Z} \cdot H, & \int_H \omega_{FS} &= \pi, & [F] = dH &\iff d = F \cdot H = \frac{1}{\pi} \int_F \omega_{FS}. \end{aligned}$$

1.1. Symplectic Surfaces.

Definition 1.1 (Symplectic surface). A submanifold $F \subseteq (M, \omega)$ is symplectic when

$$\omega|_{TF} \text{ is nondegenerate,}$$

and it is a symplectic surface when $\dim_{\mathbb{R}}(F) = 2$.

Definition 1.2 (Tame and compatible structures). An almost complex structure $J: TM \rightarrow TM$, $J^2 = -I$, is ω -tame when

$$\omega(\mathbf{v}, J\mathbf{v}) > 0 \quad (\mathbf{v} \neq \mathbf{0}),$$

and ω -compatible when additionally

$$\omega(J\mathbf{v}, J\mathbf{w}) = \omega(\mathbf{v}, \mathbf{w}).$$

Example 1.3 (Complex curves). For $F \subseteq \mathbb{C}^n$,

$$iTF = TF, \quad 0 \neq \mathbf{v} \in TF \implies i\mathbf{v} \in TF, \quad \omega_{\text{std}}(\mathbf{v}, i\mathbf{v}) = |\mathbf{v}|^2 > 0 \xrightarrow{1.1} F \text{ is symplectic.}$$

Thus smooth affine complex curves are symplectic. By Proposition 1.5, the projective case follows.

Proposition 1.4 (Affine complex submanifolds, [25]). *The standard complex structure i is ω_{std} -tame and compatible. If*

$$X = \{p_1(\mathbf{z}) = \cdots = p_N(\mathbf{z}) = 0\} \subseteq \mathbb{C}^n, \quad 0 \text{ is a regular value of } (p_1, \dots, p_N),$$

where every p_k is holomorphic, then X is symplectic.

Proof. Regularity identifies every tangent space with a complex kernel.

$$\omega_{\text{std}}(\mathbf{v}, i\mathbf{v}) = |\mathbf{v}|^2 > 0, \quad \omega_{\text{std}}(i\mathbf{v}, i\mathbf{w}) = \omega_{\text{std}}(\mathbf{v}, \mathbf{w}) \xleftrightarrow{1.2} i \text{ is tame and compatible.}$$

$$T_{\mathbf{p}}X = \ker(d(p_1, \dots, p_N)_{\mathbf{p}}), \quad dp_k \circ i = i dp_k \implies iT_{\mathbf{p}}X = T_{\mathbf{p}}X.$$

$$0 \neq \mathbf{v} \in T_{\mathbf{p}}X \implies i\mathbf{v} \in T_{\mathbf{p}}X, \quad \omega_{\text{std}}(\mathbf{v}, i\mathbf{v}) = |\mathbf{v}|^2 > 0 \xrightarrow{1.1} \omega_{\text{std}}|_{T_{\mathbf{p}}X} \text{ is nondegenerate.}$$

Hence every smooth affine complex zero set inherits ω_{std} . □

Proposition 1.5 (Fubini–Study reduction, [25]). *The diagonal S^1 -action on S^5 has quotient $\mathbb{CP}^2$, and $\iota^* \omega_{\text{std}}$ descends to a symplectic form ω_{FS} . If $P: \mathbb{C}^3 \rightarrow \mathbb{C}$ is homogeneous and $P^{-1}(0) \setminus \{\mathbf{0}\}$ is smooth, then its projectivization is symplectic.*

²Based on PCMI lectures by Laura Starkston [24].

Proof. The restricted form has precisely the orbit direction as its kernel.

$$T_{\mathbf{p}}S^5 = \{\mathbf{v} \mid g_{\text{std}}(\mathbf{p}, \mathbf{v}) = 0\}, \quad \mathbf{w} \in \ker((\iota^* \omega_{\text{std}})_{\mathbf{p}}) \iff g_{\text{std}}(i\mathbf{w}, \mathbf{v}) = 0 \quad (\mathbf{v} \perp \mathbf{p}).$$

$$\mathbf{w} \in \ker((\iota^* \omega_{\text{std}})_{\mathbf{p}}) \iff \mathbf{w} \in \mathbb{R} \cdot i\mathbf{p}.$$

$$0 \longrightarrow \mathbb{R} \cdot i\mathbf{p} \longrightarrow T_{\mathbf{p}}S^5 \xrightarrow{d\pi_{\mathbf{p}}} T_{[\mathbf{p}]} \mathbb{CP}^2 \longrightarrow 0.$$

$$(e^{i\theta})^* \iota^* \omega_{\text{std}} = \iota^* \omega_{\text{std}}, \quad \ker(\iota^* \omega_{\text{std}}) = \ker(d\pi) \implies \pi^* \omega_{FS} = \iota^* \omega_{\text{std}}.$$

$$d\omega_{FS} = 0, \quad \ker(\omega_{FS}) = \{0\}, \quad S^5/S^1 \cong \mathbb{CP}^2.$$

The horizontal complement descends the complex structure.

$$\xi_{\mathbf{p}} := (\mathbb{R}\mathbf{p} \oplus \mathbb{R}i\mathbf{p})^\perp, \quad i\xi_{\mathbf{p}} = \xi_{\mathbf{p}}, \quad d\pi_{\mathbf{p}}|_{\xi_{\mathbf{p}}} \circ i = i \circ d\pi_{\mathbf{p}}|_{\xi_{\mathbf{p}}},$$

$$0 \neq \mathbf{v} \in \xi_{\mathbf{p}} \implies \omega_{FS}(d\pi\mathbf{v}, i d\pi\mathbf{v}) = \omega_{\text{std}}(\mathbf{v}, i\mathbf{v}) = |\mathbf{v}|^2 > 0.$$

$$\omega_{FS}(id\pi\mathbf{v}, id\pi\mathbf{w}) = \omega_{\text{std}}(i\mathbf{v}, i\mathbf{w}) = \omega_{\text{std}}(\mathbf{v}, \mathbf{w}) = \omega_{FS}(d\pi\mathbf{v}, d\pi\mathbf{w}) \xrightarrow{1.2} i \text{ is } \omega_{FS}\text{-compatible.}$$

For a homogeneous polynomial P of degree d , reduction preserves its complex tangent lines.

$$P(\lambda \mathbf{z}) = \lambda^d P(\mathbf{z}), \quad dP_{\mathbf{z}} \neq 0 \quad (\mathbf{z} \in P^{-1}(0) \setminus \{0\}),$$

$$C_P := (P^{-1}(0) \cap S^5)/S^1 \text{ is smooth,} \quad \dim_{\mathbb{C}}(C_P) = 1, \quad \omega_{FS}(\mathbf{v}, i\mathbf{v}) > 0 \xrightarrow{1.1} C_P \text{ is symplectic.}$$

Thus Hopf reduction produces ω_{FS} and its smooth projective curves. $\square$

1.2. Symplectic Isotopy.

Definition 1.6 (Symplectic isotopy). Symplectic surfaces $F_0, F_1 \subseteq (M, \omega)$ are symplectically isotopic when a smooth family of embeddings $\iota_t: F \rightarrow M$ satisfies

$$\iota_t^* \omega \text{ is symplectic,} \quad \iota_0(F) = F_0, \quad \iota_1(F) = F_1.$$

Example 1.7 (Complex isotopy). For

$$\mathcal{P}_d = \mathbb{CP}^{N_d}, \quad N_d = \binom{d+2}{2} - 1, \quad \mathcal{D}_d = \{P \in \mathcal{P}_d \mid \{P=0\} \text{ is singular}\},$$

$$\text{codim}_{\mathbb{R}}(\mathcal{D}_d) \geq 2, \quad P, Q \in \mathcal{P}_d \setminus \mathcal{D}_d \implies P_t: [0, 1] \rightarrow \mathcal{P}_d \setminus \mathcal{D}_d, \quad P_0 = P, \quad P_1 = Q.$$

$$\mathcal{C} := \{(t, [\mathbf{z}]) \mid P_t(\mathbf{z}) = 0\} \subseteq [0, 1] \times \mathbb{CP}^2, \quad \rho: \mathcal{C} \rightarrow [0, 1], \quad \rho(t, [\mathbf{z}]) = t \text{ is a proper submersion,}$$

$$\rho \text{ is a smooth bundle} \implies \{P_t = 0\} \text{ is a smooth family of complex curves} \xrightarrow{1.6} \{P=0\} \simeq_{\text{symp}} \{Q=0\}.$$

Conjecture 1.8 (Symplectic isotopy). Every two embedded symplectic surfaces in $\mathbb{CP}^2$ of equal degree are symplectically isotopic.

Fact 1.9 (Known degrees, [8, 22, 20, 21]).

$$\text{Conjecture 1.8 holds for } d \leq 17, \quad d \geq 18 \text{ is open.}$$

1.3. Almost Complex Structures.

Definition 1.10 (Pseudoholomorphic curve). A map $u: (\Sigma, j) \rightarrow (M, J)$ is J -holomorphic when

$$\begin{array}{ccc} T_z \Sigma & \xrightarrow{du_z} & T_{u(z)} M \\ j_z \downarrow & & \downarrow J_{u(z)} \\ T_z \Sigma & \xrightarrow{du_z} & T_{u(z)} M. \end{array} \quad du \circ j = J \circ du,$$

Proposition 1.11 (Cauchy–Riemann equations, [25]). For

$$u = (f_1 + ig_1, f_2 + ig_2): (\mathbb{C}, i) \rightarrow (\mathbb{C}^2, i),$$

the equation $du \circ i = i \circ du$ is equivalent to

$$\partial_x f_k = \partial_y g_k, \quad \partial_y f_k = -\partial_x g_k, \quad k = 1, 2.$$

Proof. Compare the two bundle maps on ∂_x .

$$du(i\partial_x) = du(\partial_y) = \partial_y f_k + i\partial_y g_k, \quad i du(\partial_x) = -\partial_x g_k + i\partial_x f_k,$$

$$du \circ i = i \circ du \xLeftrightarrow{1.10} \partial_x f_k = \partial_y g_k, \quad \partial_y f_k = -\partial_x g_k \quad (k = 1, 2).$$

Hence the pseudoholomorphic equation is the classical Cauchy–Riemann system. $\square$

Fact 1.12 (Contractibility, [4, 14]). The spaces of ω -tame and ω -compatible almost complex structures are nonempty and contractible.

Proposition 1.13 (Compatible metrics, [25]). An almost complex structure J is ω -compatible exactly when

$$G_J(\mathbf{u}, \mathbf{v}) := \omega(\mathbf{u}, J\mathbf{v})$$

is a positive inner product. Every positive inner product G and symplectic form ω determine a compatible J .

Proof. Compatibility is equivalent to symmetry and positivity, while the converse follows from polar decomposition.

$$J \text{ compatible} \xLeftrightarrow{1.2} G_J(\mathbf{v}, \mathbf{v}) > 0, \quad G_J(\mathbf{v}, \mathbf{u}) = \omega(\mathbf{v}, J\mathbf{u}) = \omega(\mathbf{u}, J\mathbf{v}) = G_J(\mathbf{u}, \mathbf{v}).$$

$$G_J \text{ symmetric and positive} \implies \omega(J\mathbf{u}, J\mathbf{v}) = G_J(J\mathbf{u}, \mathbf{v}) = G_J(\mathbf{v}, J\mathbf{u}) = \omega(\mathbf{u}, \mathbf{v}) \xRightarrow{1.2} J \text{ compatible}.$$

$$\omega(\mathbf{u}, \mathbf{v}) = G(A\mathbf{u}, \mathbf{v}) \implies A^* = -A, \quad P := (-A^2)^{1/2} > 0, \quad AP = PA.$$

$$J := P^{-1}A, \quad J^2 = -I, \quad G_J(\mathbf{u}, \mathbf{v}) = \omega(\mathbf{u}, J\mathbf{v}) = G(P\mathbf{u}, \mathbf{v}).$$

$$P > 0 \implies G_J \text{ is symmetric and positive} \xRightarrow{1.2} J \text{ is compatible}.$$

Thus $J = P^{-1}A$ is the required compatible structure. $\square$

Proposition 1.14 (Fubini–Study charts, [25]). For

$$f(\mathbf{z}) = \log(1 + |\mathbf{z}|^2), \quad d^{\mathbb{C}}f = df \circ i,$$

the form $-dd^{\mathbb{C}}f$ is symplectic and invariant under projective chart transitions, and i is compatible with $\omega_{FS} = -\frac{1}{4}dd^{\mathbb{C}}f$.

Proof. The complex Hessian is positive, while overlapping potentials differ by a pluriharmonic term.

$$\begin{aligned} \frac{\partial^2 f}{\partial z_j \partial \bar{z}_k} &= \frac{(1 + |\mathbf{z}|^2) \delta_{jk} - \bar{z}_j z_k}{(1 + |\mathbf{z}|^2)^2}, \\ \sum_{j,k} \frac{\partial^2 f}{\partial z_j \partial \bar{z}_k} \xi_j \bar{\xi}_k &= \frac{(1 + |\mathbf{z}|^2) |\xi|^2 - |\langle \mathbf{z}, \xi \rangle|^2}{(1 + |\mathbf{z}|^2)^2} \geq \frac{|\xi|^2}{(1 + |\mathbf{z}|^2)^2} > 0. \\ \psi(z_1, z_2) &= \left(\frac{1}{z_1}, \frac{z_2}{z_1} \right), \quad f \circ \psi = f - \log |z_1|^2, \quad z_1 \neq 0, \quad \partial \bar{\partial} \log |z_1|^2 = 0. \\ \psi^*(-dd^{\mathbb{C}} f) &= -dd^{\mathbb{C}} f, \quad -dd^{\mathbb{C}} f = 2i \partial \bar{\partial} f. \end{aligned}$$

The Hopf section fixes the quotient normalization.

$$\begin{aligned} s(\mathbf{z}) &= \frac{(1, \mathbf{z})}{\sqrt{1 + |\mathbf{z}|^2}}, \quad \varphi(\mathbf{z}) = [1 : \mathbf{z}], \quad \pi \circ s = \varphi, \\ \pi^* \omega_{FS} &= \iota^* \omega_{\text{std}} \xrightarrow{1.5} \varphi^* \omega_{FS} = s^* \iota^* \omega_{\text{std}} = \frac{i}{2} \partial \bar{\partial} f = -\frac{1}{4} dd^{\mathbb{C}} f. \\ \omega_{FS}(\mathbf{v}, i\mathbf{v}) &> 0, \quad \omega_{FS}(i\mathbf{v}, i\mathbf{w}) = \omega_{FS}(\mathbf{v}, \mathbf{w}) \xrightarrow{1.2} i \text{ is } \omega_{FS}\text{-compatible}. \end{aligned}$$

Thus the local forms glue to the normalized Fubini–Study form. $\square$

Proposition 1.15 (Symplectic surfaces as J -curves, [24, 14]). *Every embedded symplectic surface $F \subseteq (M, \omega)$ is J -holomorphic for a nonempty contractible space of compatible J . Every immersed J -holomorphic curve for tame J is symplectic at its regular points.*

Proof. Choose compatible structures on the symplectic tangent and normal summands and extend the induced metric.

$$\begin{aligned} TM|_F &= TF \oplus TF^\omega, \quad J_F = J_{TF} \oplus J_{TF^\omega}, \quad g_F(\mathbf{u}, \mathbf{v}) = \omega(\mathbf{u}, J_F \mathbf{v}), \quad j := J_F|_{TF}. \\ g|_F &= g_F \xrightarrow{1.13} J|_F = J_F, \quad JTF = TF, \quad d\iota_F \circ j = J \circ d\iota_F. \\ \mathcal{J}_\omega(M; F) &:= \{J \mid JTF = TF\} \longrightarrow \mathcal{J}_\omega(TF) \times \mathcal{J}_\omega(TF^\omega), \end{aligned}$$

base and relative-extension fibers are nonempty and contractible $\xrightarrow{1.12} \mathcal{J}_\omega(M; F)$ is nonempty and contractible.

$$\begin{aligned} du \circ j &= J \circ du \implies J(du(T\Sigma)) = du(T\Sigma), \\ 0 \neq \mathbf{v} \in du(T\Sigma), \quad \omega(\mathbf{v}, J\mathbf{v}) &> 0 \xrightarrow{1.2} \omega|_{du(T\Sigma)} \text{ is nondegenerate}. \end{aligned}$$

Thus symplectic surfaces and regular tame J -curves agree after choosing J . $\square$

1.4. Adjunction and Intersection.

Definition 1.16 (Chern and normal data, [24]). For an embedded J -holomorphic curve $F \subseteq M$,

$$c_1(J) := c_1(TM, J), \quad NF := TM|_F / TF, \quad \langle c_1(NF), [F] \rangle = [F]^2.$$

Theorem 1.17 (Adjunction, [13]). *If $F \subseteq M$ is a connected embedded J -holomorphic curve of genus g , then*

$$\langle c_1(J), [F] \rangle = 2 - 2g + [F]^2.$$

Proof. The complex normal sequence splits the Chern number into tangent and normal terms.

$$\begin{aligned}
0 &\longrightarrow TF \longrightarrow TM|_F \longrightarrow NF \longrightarrow 0 \\
c_1(TM|_F) &= c_1(TF) + c_1(NF). \\
\langle c_1(TF), [F] \rangle &= \chi(F) = 2 - 2g. \\
\langle c_1(NF), [F] \rangle &= [F]^2. \\
\langle c_1(J), [F] \rangle &= 2 - 2g + [F]^2.
\end{aligned}$$

This is the adjunction identity. □

Example 1.18 (Degree and genus, [24]). For $[F] = dH \in H_2(\mathbb{CP}^2; \mathbb{Z})$,

$$\begin{aligned}
H^2 &= 1, & \langle c_1(T\mathbb{CP}^2), H \rangle &= 3, \\
3d &= 2 - 2g + d^2 \xLeftrightarrow{1.17} g = \frac{(d-1)(d-2)}{2}.
\end{aligned}$$

Proposition 1.19 (Blow-up adjunction, [25]). *For the standard symplectic blow-up*

$$\begin{aligned}
X &= \mathbb{CP}^2 \# n \overline{\mathbb{CP}}^2, & n &\geq 3, & c_1 &= 3\text{PD}(h) - \sum_{i=1}^n \text{PD}(e_i), \\
\langle c_1, h \rangle &= 3, & \langle c_1, e_i \rangle &= 1.
\end{aligned}$$

The class $3h - 2e_1 - e_2 - e_3$ has genus 0, while $3h - 5e_1$ and $-3e_1 - e_2 - e_3$ have no connected embedded symplectic representatives.

Proof. The diagonal intersection form and adjunction give the three calculations.

$$\begin{aligned}
h^2 &= 1, & e_i^2 &= -1, & h \cdot e_i &= e_i \cdot e_j = 0 \quad (i \neq j). \\
\left\langle c_1, a_0 h - \sum_i a_i e_i \right\rangle &= 3a_0 - \sum_i a_i. \\
A_1 &:= 3h - 2e_1 - e_2 - e_3, & \langle c_1, A_1 \rangle &= 5, & A_1^2 &= 3 \xrightarrow{1.17} g(A_1) = 0. \\
A_2 &:= 3h - 5e_1, & \langle c_1, A_2 \rangle &= 4, & A_2^2 &= -16 \xrightarrow{1.17} g(A_2) = -9. \\
A_3 &:= -3e_1 - e_2 - e_3, & \langle c_1, A_3 \rangle &= -5, & A_3^2 &= -11 \xrightarrow{1.17} g(A_3) = -2.
\end{aligned}$$

The negative formal genera obstruct the asserted representatives. □

Proposition 1.20 (Positivity of intersection, [13]). *For an ω -tame J , closed simple J -holomorphic curves with distinct images and no common component have isolated intersections of positive local multiplicity.*

Proof. At a transverse intersection the two complex tangent lines give the ambient orientation.

$$\begin{aligned}
0 \neq \mathbf{v} &\in T_p C_1, & 0 \neq \mathbf{w} &\in T_p C_2, & T_p C_1 \cap T_p C_2 &= \{\mathbf{0}\}. \\
(\mathbf{v}, J\mathbf{v}, \mathbf{w}, J\mathbf{w}) \text{ positive} &\xrightarrow{1.2} i_p(C_1, C_2) = 1. \\
p \text{ nontransverse} &\implies i_p(C_1, C_2) \in \mathbb{Z}_{>0}. \\
[C_1] \cdot [C_2] &= \sum_{p \in C_1 \cap C_2} i_p(C_1, C_2) \geq 0.
\end{aligned}$$

Thus every local contribution is positive. □

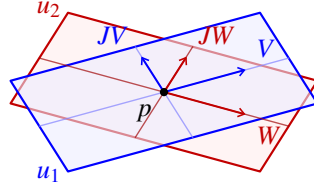


FIGURE 1. A schematic projection of the transverse J -complex tangent planes at p ; their oriented axes give the positive basis (V, JV, W, JW) .

1.5. Degree-One Isotopy.

Definition 1.21 (Spherical moduli and evaluation, [13, 24]). For $A \in H_2(M; \mathbb{Z})$, set

$$\mathcal{M}_p(J, A) := \{u: S^2 \rightarrow M \mid du \circ i = J \circ du, [u] = A, u \text{ simple}\},$$

$$\mathcal{M}_2(J, A) := \frac{\mathcal{M}_p(J, A) \times ((S^2 \times S^2) \setminus \Delta_{S^2})}{\text{PSL}(2, \mathbb{C})},$$

$$\text{ev}_J: \mathcal{M}_2(J, A) \rightarrow M \times M, \quad \text{ev}_J([u, z_1, z_2]) = (u(z_1), u(z_2)).$$

Write $\overline{\mathcal{M}}_2(J, A)$ for the stable-map compactification.

Theorem 1.22 (A unique J -line through two points, [8]). *For every ω_{FS} -tame J and distinct $p, q \in \mathbb{CP}^2$, exactly one embedded J -holomorphic sphere in class H contains p, q .*

Proof. Primitivity, minimal area, and generalized adjunction give a compact regular moduli problem.

$$H \text{ primitive} \implies \text{no multiple cover represents } H.$$

$$A_v = d_v H, \quad d_v = \frac{\int_{A_v} \omega_{FS}}{\int_H \omega_{FS}} \in \mathbb{Z}_{>0}, \quad \sum_{v=1}^k d_v = 1 \implies k = 1.$$

$$\text{two marks, } u(z_1) \neq u(z_2) \implies \text{no stable ghost tree occurs.}$$

$$2g(S^2) - 2 = H^2 - \langle c_1, H \rangle - 2\delta(u) = -2 - 2\delta(u) \xrightarrow{[12]} \delta(u) = 0.$$

$$c_1(N_u) = H^2 = 1, \quad c_1(N_u(-z_1 - z_2)) = -1 > -2 \xrightarrow{[11, \text{Section 3}]} D_u^N, D_{u, z_1, z_2}^N \text{ surjective.}$$

$$\dim \mathcal{M}_p(J, H) = 10, \quad \dim \mathcal{M}(J, H) = 4, \quad \dim \mathcal{M}_2(J, H) = 8.$$

The evaluation degree is preserved along a tame path from J to i .

$$U := (\mathbb{CP}^2 \times \mathbb{CP}^2) \setminus \Delta_{\mathbb{CP}^2}, \quad \mathcal{M}_2^U(J, H) := \text{ev}_J^{-1}(U), \quad \text{ev}_J^U := \text{ev}_J|_{\mathcal{M}_2^U(J, H)}.$$

$$H \text{ minimal, } (u(z_1), u(z_2)) \in U \implies \overline{\mathcal{M}}_2(J, H)|_U = \mathcal{M}_2^U(J, H), \quad \text{ev}_J^U \text{ proper.}$$

$$J_t: J_0 = J \simeq J_1 = i, \quad \mathcal{W}_2^U(H) := \bigcup_{t \in [0, 1]} \{t\} \times \mathcal{M}_2^U(J_t, H),$$

$$\text{Ev}^U(t, x) := (t, \text{ev}_{J_t}^U(x)), \quad \mathfrak{t}_s(x) := (s, x).$$

$$\mathcal{M}_2^U(J, H) \longleftarrow \mathcal{W}_2^U(H) \longleftarrow \mathcal{M}_2^U(i, H)$$

$$\begin{array}{ccccc} \text{ev}_J^U \downarrow & & \downarrow \text{Ev}^U & & \downarrow \text{ev}_i^U \\ U & \xleftarrow{\mathfrak{t}_0} & \times U & \xleftarrow{\mathfrak{t}_1} & U \end{array}$$

$$\text{Ev}^U \text{ proper, } \partial \mathcal{W}_2^U(H) = \mathcal{M}_2^U(i, H) - \mathcal{M}_2^U(J, H).$$

$$\#_{\text{or}}(\text{ev}_i^U)^{-1}(p, q) = 1 \implies \deg(\text{ev}_J^U) = 1 \implies (\text{ev}_J^U)^{-1}(p, q) \neq \emptyset.$$

$$C_1, C_2 \ni p, q, \quad C_1 \neq C_2 \xrightarrow{1.20} 1 = H \cdot H \geq i_p(C_1, C_2) + i_q(C_1, C_2) \geq 2.$$

Thus the J -line exists and is unique. $\square$

Theorem 1.23 (Gromov's degree-one isotopy, [8]). *Every embedded symplectic surface $\Sigma \subseteq (\mathbb{CP}^2, \omega_{FS})$ in class H is symplectically isotopic to a complex line.*

Proof. Choose J_1 making Σ holomorphic and a compatible path from i to J_1 .

$$\Sigma \text{ symplectic} \xrightarrow{1.15} \Sigma \text{ } J_1\text{-holomorphic.}$$

$$J_t: J_0 = i \simeq J_1, \quad p \neq q \in \Sigma.$$

$$\Sigma_t := \text{the unique } J_t\text{-line through } p, q \xrightarrow{1.22} \Sigma_0 \text{ complex,} \quad \Sigma_1 = \Sigma.$$

$$\mathcal{J}_{p,q} := \{(t, C) \mid C \text{ a } J_t\text{-line through } p, q\}, \quad \rho(t, C) := t.$$

$$c_1(N_{\Sigma_t}(-p-q)) = -1 > -2 \xrightarrow{[11, \text{Section 3}]} D_{\Sigma_t, p, q}^N \text{ surjective.}$$

Automatic regularity and minimal area identify the universal moduli with the parameter interval.

$$D_{\Sigma_t, p, q}^N \text{ surjective,} \quad H \text{ minimal} \implies \rho: \mathcal{J}_{p,q} \rightarrow [0, 1] \text{ proper local diffeomorphism.}$$

$$\# \rho^{-1}(t) = 1 \xrightarrow{1.22} \rho \text{ a diffeomorphism} \implies t \mapsto \Sigma_t \text{ smooth.}$$

$$\mathcal{U} := \{(t, x) \mid x \in \Sigma_t\}, \quad \mathcal{U} \rightarrow [0, 1] \text{ a proper submersion} \implies \mathcal{U} \cong [0, 1] \times \Sigma_0,$$

$$\Theta(t, x) = (t, \iota_t(x)) \implies \iota_t: \Sigma_0 \hookrightarrow \mathbb{CP}^2, \quad \iota_t(\Sigma_0) = \Sigma_t.$$

$$J_t T\Sigma_t = T\Sigma_t, \quad \omega_{FS}(\mathbf{v}, J_t \mathbf{v}) > 0 \xrightarrow{1.1} \omega_{FS}|_{T\Sigma_t} \text{ nondegenerate.}$$

$$\Sigma_0 \simeq_{\text{symp}} \Sigma_1 \xleftarrow{1.6} \mathbb{CP}^1 \simeq_{\text{symp}} \Sigma.$$

Thus Σ_t is the required symplectic isotopy. $\square$

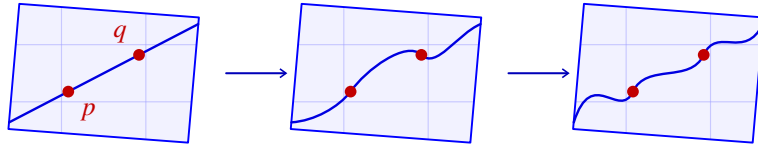


FIGURE 2. The family of unique J_t -lines through the fixed points p, q , from the complex line at $t = 0$ to Σ at $t = 1$.

Proposition 1.24 (Five points determine at most one conic, [25]). *At most one smooth embedded degree-two J -holomorphic curve passes through five fixed distinct points.*

Proof. Suppose two distinct embedded conics C_1, C_2 contain $p_1, \dots, p_5$.

$$[C_1] = [C_2] = 2H, \quad C_1 \cdot C_2 = 4.$$

$$i_{p_k}(C_1, C_2) \geq 1 \quad (1 \leq k \leq 5).$$

$$4 = (2H) \cdot (2H) = \sum_{p \in C_1 \cap C_2} i_p(C_1, C_2) \geq \sum_{k=1}^5 i_{p_k}(C_1, C_2) \geq 5.$$

This contradiction proves uniqueness. $\square$

Remark 1.25 (Higher degree, [24]). For $d > 1$, cusp degeneration and multiple covers obstruct the degree-one continuation argument.

2. SINGULAR AND BRAIDED SURFACES

We pass from local singularities to resolution, isotopy, braid monodromy, and wiring diagrams [24].

$$B_\varepsilon^4(p) \subseteq M, \quad S_\varepsilon^3(p) = \partial B_\varepsilon^4(p), \quad K_p := \Sigma \cap S_\varepsilon^3(p), \quad H = [\mathbb{CP}^1] \in H_2(\mathbb{CP}^2; \mathbb{Z}).$$

$$[\Sigma] = dH, \quad \pi_p: \mathbb{CP}^2 \setminus \{p\} \rightarrow \mathbb{CP}^1, \quad B_d = d\text{-strand braid group}, \quad \Delta_d^2 = \text{full twist}.$$

2.1. Singularities and Resolution.

Definition 2.1 (Singular symplectic surface). A surface $\Sigma \subseteq (M, \omega)$ is singular symplectic when its singular set is finite, every tangent plane on the smooth locus is symplectic, and local symplectic coordinates identify each singular pair with a complex plane-curve singularity; K_p is its *link*, and such links are *algebraic links*.

Example 2.2 (Transverse cones). Every positive transverse link has a symplectic cone away from its vertex.

$$\begin{aligned} K \subseteq S^3, \quad \text{Cone}_\varepsilon(K) &:= \{r\mathbf{x} \mid \mathbf{x} \in K, 0 \leq r \leq \varepsilon\} \subseteq B_\varepsilon^4, \quad \text{Cone}_\varepsilon(K) \cap S_r^3 = rK. \\ \lambda_{\text{std}} &:= \frac{1}{2} \sum_{k=1}^2 (x_k dy_k - y_k dx_k), \quad \alpha := \lambda_{\text{std}}|_{S^3}, \quad F(r, t) := r\gamma(t), \quad F^*\lambda_{\text{std}} = r^2\alpha(\dot{\gamma}(t))dt. \\ \gamma \text{ positively transverse} &\implies F^*\omega_{\text{std}} = d(F^*\lambda_{\text{std}}) = 2r\alpha(\dot{\gamma}(t))dr \wedge dt > 0 \quad (r > 0). \end{aligned}$$

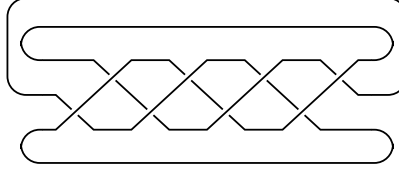


FIGURE 3. The link of $x^3 = y^4$ is the $(3, 4)$ -torus knot.

Example 2.3 (Torus-knot links, [26]). For relatively prime $p, q > 0$, the Milnor link of $C_{p,q} = \{x^p = y^q\}$ is isotopic to the closed braid

$$C_{p,q} \cap (S_\varepsilon^1 \times \mathbb{C}) = \left\{ \left(\varepsilon e^{i\theta}, \varepsilon^{p/q} e^{i(p\theta + 2\pi k)/q} \right) \mid 0 \leq k < q, 0 \leq \theta < 2\pi \right\}.$$

$$\theta: 0 \longrightarrow 2\pi \implies q \text{ points rotate through } 2\pi p/q \implies K_0(C_{p,q}) = T(p, q).$$

Thus this branch gives a torus knot; a general irreducible branch gives an iterated torus knot.

Definition 2.4 (Blow-up and transforms). The local blow-up and its exceptional divisor are

$$B := \{((x, y), [u : v]) \in \mathbb{C}^2 \times \mathbb{CP}^1 \mid xv - yu = 0\}, \quad \pi((x, y), [u : v]) = (x, y), \quad E := \pi^{-1}(0) \cong \mathbb{CP}^1,$$

$$\begin{array}{ccccc} E & \xrightarrow{\quad} & B & \xleftarrow{\quad} & B \setminus E \\ \downarrow & & \downarrow \pi & & \downarrow \cong \\ \{0\} & \xrightarrow{\quad} & \mathbb{C}^2 & \xleftarrow{\quad} & \mathbb{C}^2 \setminus \{0\} \end{array}$$

For a curve C of multiplicity m in a global surface,

$$C^{\text{tot}} := \pi^*C = \tilde{C} + mE, \quad \tilde{C} := \overline{\pi^{-1}(C \setminus \{0\})}, \quad [\tilde{C}] = \pi^*[C] - m[E], \quad \tilde{C}^2 = C^2 - m^2, \quad \tilde{C} \cdot E = m.$$

Proposition 2.5 (Embedded cusp resolutions, [25]). *The embedded resolutions satisfy*

$$\begin{aligned} x^3 = y^2: \quad \mathbf{m}_{>1} = [2], \quad \#B1 = 3, \quad \mathbf{m}_E = (2, 3, 6), \\ x^p = y^{p-1}: \quad \mathbf{m}_{>1} = [p-1], \quad \#B1 = p, \quad x^3 = y^5: \quad \mathbf{m}_{>1} = [3, 2], \quad \#B1 = 4, \quad \mathbf{m}_E = (3, 5, 9, 15). \end{aligned}$$

Here $p \geq 3$, and SNC means that at most two local components meet.

Proof. The affine charts separate singular multiplicity from normal-crossing blow-ups.

$$\begin{aligned} u = 1, \quad y = xv, \quad x^3 - y^2 = x^2(x - v^2) \xrightarrow{2.4} \pi^*C = 2E_1 + \tilde{C}, \quad E_1 = \{x = 0\}, \quad \tilde{C} = \{x = v^2\}. \\ T_0E_1 = \{x = 0\} = T_0\tilde{C}, \quad x = vs, \quad x = v^2 \implies v^3s^2(s - v) = 0. \\ E'_1 = \{s = 0\}, \quad E_2 = \{v = 0\}, \quad \tilde{C} = \{s - v = 0\} \implies [0 : 1], [1 : 0], [1 : 1] \in E_3 \cong \mathbb{CP}^1. \\ \{x^p = y^{p-1}\}, \quad y = xv \implies x^{p-1}(x - v^{p-1}) = 0, \quad \tilde{C}_1 = \{x = v^{p-1}\}. \end{aligned}$$

The remaining tangency exponent drops by one at each blow-up, followed by one blow-up of the terminal triple point.

$$\begin{aligned} \tilde{C}_k = \{x_k = v^{p-k}\} \quad (1 \leq k \leq p-1) \implies \#B1 = p, \quad \mathbf{m}_{>1} = [p-1]. \\ x^3 - y^5, \quad x = yu \implies y^3(u^3 - y^2) = 0 \xrightarrow{2.4} \{u^3 = y^2\}. \\ \mathbf{m}_{>1} = [3, 2], \quad \mathbf{m}_E = (3, 5, 9, 15), \quad \#B1 = 1 + 3 = 4. \end{aligned}$$

Thus the displayed sequences record singular centers, while the stated counts give simple normal crossings. $\square$

Theorem 2.6 (McDuff–Micallef–White local model, [12, 15]). *The image germ of a somewhere-injective J -curve at an isolated singular point is C^1 -equivalent to a reduced complex plane-curve germ; hence its oriented link is isotopic to an algebraic link.*

2.2. Equisingular Isotopy.

Definition 2.7 (Equisingular symplectic isotopy). Singular symplectic surfaces Σ_0, Σ_1 are equisingularly symplectically isotopic when a smooth family Σ_t , with marked singular points p_t^j , joins them and satisfies

$$\Sigma_t \setminus \{p_t^j\}_j \text{ smooth}, \quad K_{p_t^j} \cong K_{p_0^j} \quad (0 \leq t \leq 1).$$

Definition 2.8 (Symplectic line arrangement). A symplectic line arrangement is a union $L = \bigcup_{j=1}^n L_j \subseteq \mathbb{CP}^2$ of embedded symplectic spheres in class H , each pair meeting once positively; it is generic when every singularity is a transverse double point.

Theorem 2.9 (Barraud’s line-arrangement isotopy, [2, 23]). *Every generic symplectic line arrangement in $\mathbb{CP}^2$ is equisingularly symplectically isotopic to a complex line arrangement.*

Proof. Choose generic moving two-point constraints along a generic path of compatible structures.

$$L_j \cdot L_k = 1 > 0, \quad \text{simultaneous local complex models} \xrightarrow{1.15} J_0 TL_j = TL_j \quad (1 \leq j \leq n).$$

$$J_t: J_0 \simeq J_1 = i, \quad p_j(t) \neq q_j(t), \quad L_j^t := J_t\text{-line through } p_j(t), q_j(t) \xrightarrow{1.22} L^t := \bigcup_j L_j^t.$$

$$\mathcal{T}_{abc} := \{L_a^t \cap L_b^t \cap L_c^t \neq \emptyset\}, \quad \text{codim}_{\mathbb{R}} \mathcal{T}_{abc} = 2 > \dim[0, 1] \implies L^t \text{ has no triple point.}$$

$$L_j^t \neq L_k^t \xrightarrow{1.20} \#(L_j^t \cap L_k^t) = L_j^t \cdot L_k^t = H^2 = 1, \quad L^1 \text{ complex and generic.}$$

Thus the path is an equisingular symplectic isotopy. $\square$

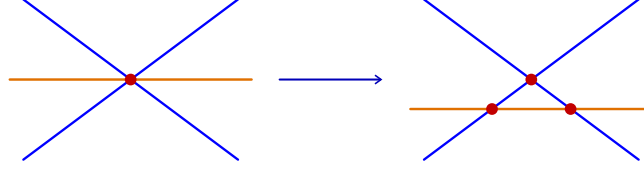


FIGURE 4. A small perturbation of one J' -line resolves a triple point into three transverse double points.

Example 2.10 (Rational cuspidal model). For $\gcd(p, q) = 1$,

$$\delta(T(p, q)) = \frac{(p-1)(q-1)}{2}, \quad g_a(d) = \frac{(d-1)(d-2)}{2}, \quad \delta(T(p, p-1)) = g_a(p)$$

$$g_a(p) - \delta(T(p, p-1)) = 0 \xrightarrow{1.17} \{x^p = y^{p-1}z\} \subseteq \mathbb{CP}^2 \quad \text{is rational with one } (p, p-1)\text{-cusp.}$$

Theorem 2.11 (McDuff's positive-sphere theorem, [10]). *If a closed connected symplectic four-manifold (V, ω) contains an embedded symplectic sphere S with $S^2 = 1$, then (V, ω) is a symplectic blow-up of $\mathbb{CP}^2$, with S sent to a line disjoint from the blow-up points.*

Theorem 2.12 (Golla–Starkston rational-cuspidal isotopy, [7]). *A degree- p singular symplectic surface in $\mathbb{CP}^2$ with one $(p, p-1)$ -cusp is rational and has a unique equisingular symplectic isotopy class, represented by a complex curve.*

Proof. Since the cusp accounts for the full arithmetic genus, the normalization is rational; the birational reduction of [7] is uniform in p , and the cubic movie below displays its local pattern.

$$g(\Sigma^v) = g_a(p) - \delta(T(p, p-1)) = \frac{(p-1)(p-2)}{2} - \frac{(p-1)(p-2)}{2} = 0.$$

$$[\Sigma] = pH, \quad [\tilde{\Sigma}] = pH - (p-1)e_1 - \sum_{i=2}^{2p-1} e_i, \quad \tilde{\Sigma}^2 = p^2 - (p-1)^2 - (2p-2) = 1.$$

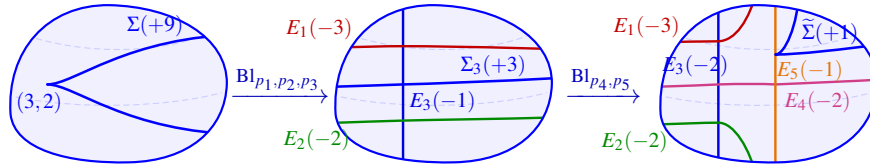


FIGURE 5. Local incidence charts for the rational $(3, 2)$ -cuspidal cubic, its resolution after three blow-ups, and the square-one configuration after two further blow-ups; crossings in the last two panels are transverse.

$$p = 3 : \quad 1 = \tilde{\Sigma}^2 \xrightarrow{2.11, [7, \S 6.1]} \begin{pmatrix} E_1(-3) & - & E_3(-2) & - & E_4(-2) & - & E_5(-1) & - & \tilde{\Sigma}(+1) \\ & & | & & & & & & \\ & & E_2(-2) & & & & & & \end{pmatrix}.$$

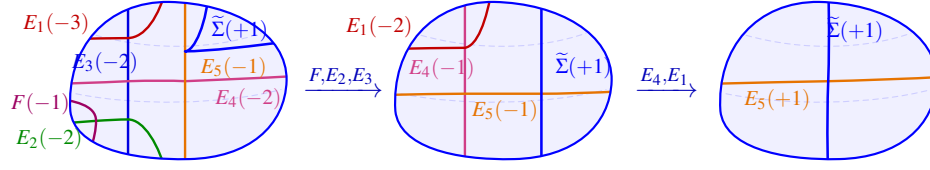


FIGURE 6. Blowing down F, E_2, E_3 and then E_4, E_1 turns $\tilde{\Sigma} \cup E_5$ into two transverse symplectic lines.

$$\tilde{\Sigma} \cup E_5 \xrightarrow{2.9} \text{two complex lines} \xrightarrow{[7, \text{Theorem 6.5}]} \Sigma \simeq_{\text{eq.symp}} \{x^p = y^{p-1}z\}.$$

□

2.3. Branched Projections and Braid Monodromy.

Definition 2.13 (Branched covering). A map $f: \Sigma_1 \rightarrow \Sigma_2$ is a branched covering when it is a covering away from finitely many points and, in local complex coordinates, has the form

$$f(z) = z^k, \quad k \in \mathbb{Z}_{>0};$$

it is simple when every nontrivial local degree is $k = 2$.

Proposition 2.14 (The J -line pencil, [24]). *Let $\Sigma \subseteq \mathbb{CP}^2$ be a degree- d symplectic surface, choose a compatible J making Σ holomorphic, a J -line L , and a generic $p \notin L \cup \Sigma$. The J -line pencil defines*

$$\pi_p: \mathbb{CP}^2 \setminus \{p\} \rightarrow L \cong \mathbb{CP}^1$$

whose restriction to Σ is a simple degree- d branched covering.

Proof. The moduli of J -lines through p foliates the complement of the center.

$$q \in L, \quad F_q := \text{unique } J\text{-line through } p, q \xrightarrow{1.22} \mathbb{CP}^2 \setminus \{p\} = \bigsqcup_{q \in L} (F_q \setminus \{p\}).$$

$$\begin{array}{ccccc} \Sigma & \hookrightarrow & \mathbb{CP}^2 \setminus \{p\} & \longleftarrow & F_q \\ \pi_p|_{\Sigma} \downarrow & & \downarrow \pi_p & & \downarrow \\ L & \xlongequal{\quad} & L & \longleftarrow & q \end{array}$$

$$F_q \cdot \Sigma = H \cdot (dH) = d \xrightarrow{1.20} \#(F_q \cap \Sigma) = d \text{ with multiplicity.}$$

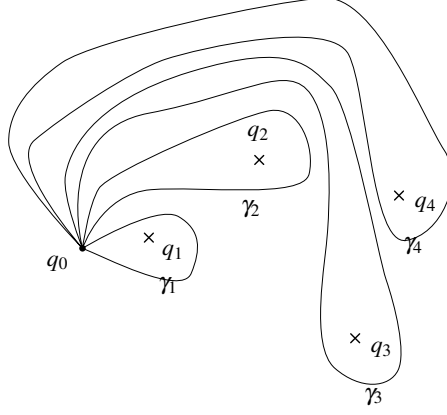
$$I_0(F_0, \Sigma) = k \xrightarrow{2.6} \pi_p|_{\Sigma}(w) = w^k, \quad p \text{ generic} \implies k = 2.$$

Thus the restricted pencil is simple of degree d .

□

Fix the critical values, a reference fiber, and the geometric basis

$$q_1, \dots, q_N, \quad q_{\infty}, q_0 \notin \{q_i\}, \quad \gamma_i \subseteq \mathbb{CP}^1 \setminus \{q_{\infty}, q_1, \dots, q_N\}, \quad \gamma_i \text{ encloses } q_i.$$

FIGURE 7. A geometric basis in $\mathbb{CP}^1 \setminus \{q_\infty, q_1, \dots, q_N\}$.

Proposition 2.15 (Braid monodromy, [19, 1, 5]). *After identifying the reference fiber with a disk containing d marked points, parallel transport defines*

$$\Phi: \pi_1(\mathbb{CP}^1 \setminus \{q_\infty, q_1, \dots, q_N\}, q_0) \rightarrow B_d, \quad \Phi([\gamma]) = [\Sigma \cap \pi_p^{-1}(\gamma)],$$

and, reading products left to right, $\beta_1 \cdots \beta_N = \Delta_d^2$, up to global conjugation and Hurwitz moves.

Proof. Trivialize the pencil over each based loop.

$$\gamma \subseteq \mathbb{CP}^1 \setminus \{q_\infty, q_i\} \longmapsto \pi_p^{-1}(\gamma) \cong \mathbb{C} \times S^1 \longmapsto \beta_\gamma := \Sigma \cap \pi_p^{-1}(\gamma) \longmapsto [\beta_\gamma] \in B_d$$

$$\Sigma \cdot F_q = d \implies \beta_\gamma \text{ has } d \text{ strands}, \quad \beta_{\gamma\eta} = \beta_\gamma \beta_\eta \implies \Phi \text{ is a homomorphism.}$$

$$z_1 = z_2^2 \implies \beta_{\partial D} = \text{positive half twist}, \quad \Delta_n = \text{positive Garside half twist.}$$

$$n \text{ transverse branches} \implies \beta_{\partial D} = \Delta_n^2.$$

$$\text{base-path change} \implies \text{conjugation}, \quad (\beta_i, \beta_{i+1}) \longmapsto (\beta_{i+1}, \beta_{i+1}^{-1} \beta_i \beta_{i+1}).$$

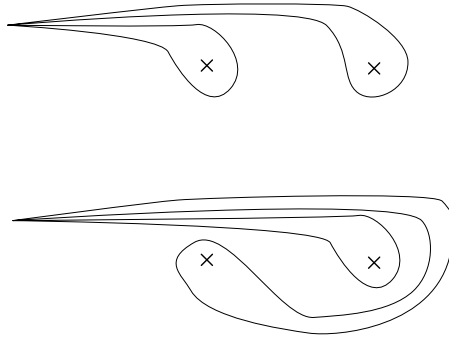


FIGURE 8. A Hurwitz move on adjacent factors.

$$\gamma_1 \cdots \gamma_N \simeq \partial D_\infty, \quad \beta_{\partial D_\infty} = \Delta_d^2 \implies \beta_1 \cdots \beta_N = \Delta_d^2.$$

Thus the factorization and local models determine the braided surface up to fiber-preserving isotopy. $\square$

2.4. Braided Wiring Diagrams.

Definition 2.16 (Braided wiring diagram, [19, 1, 5, 3]). A braided wiring diagram is the singular d -strand braid $\Sigma \cap \pi_p^{-1}(\eta)$ over an oriented path $\eta \subseteq \mathbb{CP}^1$ through all critical values, with every singular point labeled by its local model.

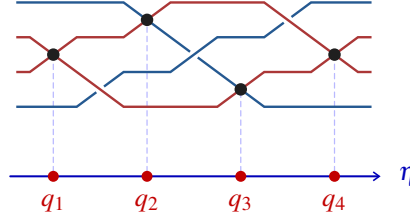


FIGURE 9. A four-strand braided wiring diagram over η ; ordinary white-gap crossings record braid motion, while the black vertices lie over the critical values $q_1, \dots, q_4$.

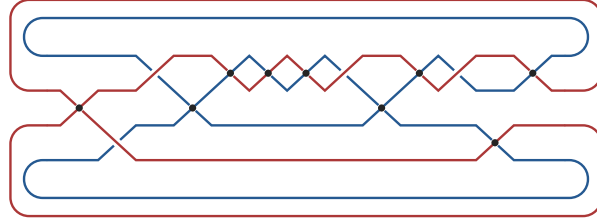


FIGURE 10. An example of a braided wiring diagram.

Proposition 2.17 (Pushoff formula, [3]). If β_+ and β_- are the smooth braids over the upper and lower pushoffs of a wiring path, read from left to right, then the monodromy around the wiring diagram is

$$\beta_{\partial} = \beta_+^{-1} \beta_-.$$

Proof. The perturbed tangency model separates the upper and lower local contributions.

$$e^{i\mu} z_1 = z_2^2, \quad z_1 = \varepsilon e^{i\theta} \implies z_2 = \pm \sqrt{\varepsilon} e^{i(\theta+\mu)/2}, \quad 0 < \mu \ll \frac{\pi}{2}.$$

$$\theta: 0 \longrightarrow 2\pi \implies z_2 \longmapsto -z_2 \implies \text{one positive half twist}, \quad \beta_+ = \sigma^{-1}, \quad \beta_- = 1, \quad \beta_+^{-1} \beta_- = \sigma.$$

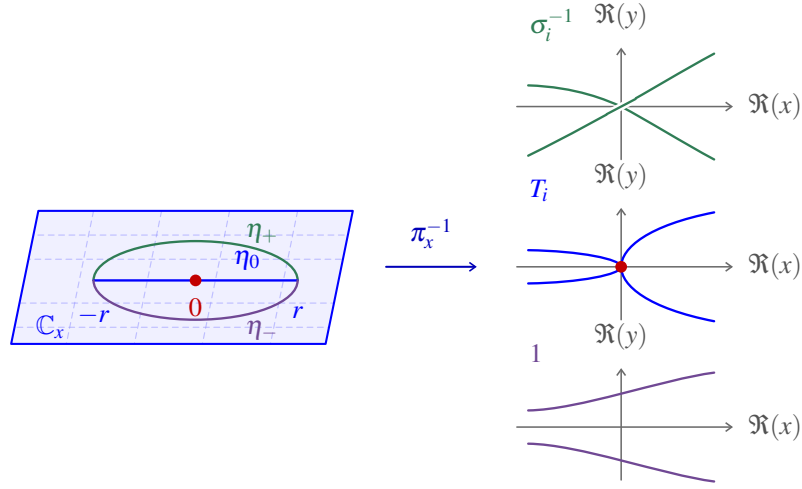


FIGURE 11. For the tangency model $y^2 = e^{i\mu}x$, the upper, real, and lower paths lift respectively to the braid σ_i^{-1} , the singular tangency T_i , and the trivial braid.

$$\begin{aligned} \text{transverse } n\text{-fold point} &\implies \beta_+ = \Delta_n^{-1}, \quad \beta_- = \Delta_n. \\ \text{upper boundary traversed right to left} &\implies \beta_+^{-1} \beta_- = \Delta_n^2. \end{aligned}$$

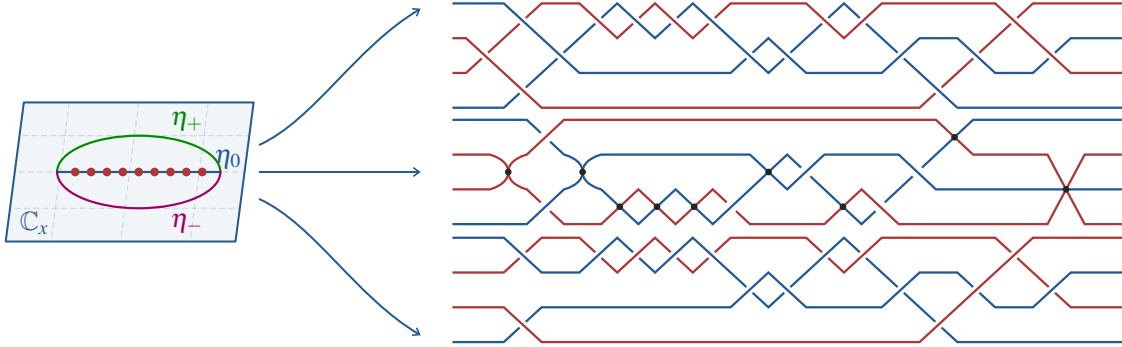


FIGURE 12. Upper and lower pushoffs of an example of a braided wiring diagram.

$$\partial N(\eta) = \eta_+^{-1} \eta_- \implies \Phi(\partial N(\eta)) = \Phi(\eta_+)^{-1} \Phi(\eta_-) = \beta_+^{-1} \beta_-.$$

Hence the two pushoffs recover the boundary braid. $\square$

Part 2. Exercises

3. EXERCISES

Exercise 3.1 (1.14). Show

$$i \text{ is } \omega_{\text{std}}\text{-compatible on } \mathbb{C}^2, \quad i \text{ is } \omega_{FS}\text{-compatible on } \mathbb{CP}^2.$$

Exercise 3.2 (1.4). For $z_j = x_j + iy_j$, let

$$\omega_{\text{std}} = \sum_{j=1}^n dx_j \wedge dy_j, \quad i\partial_{x_j} = \partial_{y_j}, \quad i\partial_{y_j} = -\partial_{x_j}.$$

(1) Show that i is ω_{std} -tame and compatible.

(2) If $P = (p_1, \dots, p_N): \mathbb{C}^n \rightarrow \mathbb{C}^N$ is holomorphic and $\mathbf{0}$ is a regular value, show that $P^{-1}(\mathbf{0})$ is symplectic.

Exercise 3.3 (1.5). Let

$$S^5 = \left\{ \mathbf{z} \in \mathbb{C}^3 \mid \sum_{j=1}^3 |z_j|^2 = 1 \right\}, \quad e^{i\theta} \cdot \mathbf{z} = e^{i\theta} \mathbf{z}, \quad \omega_{\text{std}} = \sum_{j=1}^3 dx_j \wedge dy_j.$$

(1) Show that the action preserves S^5 and $S^5/S^1 \cong \mathbb{CP}^2$.

(2) Compute $T_{\mathbf{z}}S^5$ and

$$\ker(\omega_{\text{std}}|_{T_{\mathbf{z}}S^5}).$$

(3) Show that the orbit direction generates this kernel and that $\omega_{\text{std}}|_{T_{\mathbf{z}}S^5}$ descends to a symplectic form on $\mathbb{CP}^2$.

(4) For $p = z_1^2 + z_3^2 + z_1 z_2$, show that

$$(p^{-1}(0) \cap S^5)/S^1 \subseteq \mathbb{CP}^2$$

is symplectic.

Exercise 3.4 (1.13). Let (V, ω) be symplectic.

(1) Show that J is ω -compatible exactly when

$$G_J(\mathbf{u}, \mathbf{v}) := \omega(\mathbf{u}, J\mathbf{v})$$

is a positive inner product.

(2) Given a positive inner product G , define

$$\tilde{\omega}(\mathbf{u}) = \omega(\mathbf{u}, \cdot), \quad \tilde{G}(\mathbf{u}) = G(\mathbf{u}, \cdot).$$

(a) Show that $\tilde{\omega}$ and $\tilde{G}$ are isomorphisms and obtain $A: V \rightarrow V$ satisfying

$$\omega(\mathbf{u}, \mathbf{v}) = G(A\mathbf{u}, \mathbf{v}).$$

(b) Show $A^* = -A$.

(c) Show

$$(A \cdot A^*)^* = A \cdot A^*.$$

(d) Show

$$G((A \cdot A^*)\mathbf{u}, \mathbf{u}) > 0 \quad (\mathbf{u} \neq \mathbf{0}).$$

(e) Show that

$$-A^2 = A \cdot A^*$$

has a positive symmetric square root $P = \sqrt{-A^2}$.

(f) For $J = P^{-1} \cdot A$, show

$$J \cdot J^* = I, \quad J^* = -J, \quad J^2 = -I.$$

(g) Show

$$\omega(\mathbf{v}, J\mathbf{v}) > 0 \quad (\mathbf{v} \neq \mathbf{0}).$$

(h) Show

$$\omega(J\mathbf{v}, J\mathbf{w}) = \omega(\mathbf{v}, \mathbf{w}).$$

Exercise 3.5 (1.14). Let

$$d^{\mathbb{C}}f = df \circ i, \quad f(z_1, z_2) = \log(1 + |z_1|^2 + |z_2|^2).$$

(1) Show that f is strictly plurisubharmonic and

$$\omega_{FS} := -dd^{\mathbb{C}}f$$

is symplectic.

(2) Show that

$$\mathcal{U}_j = \{[z_0 : z_1 : z_2] \in \mathbb{CP}^2 \mid z_j \neq 0\}$$

with

$$\phi_0 = \left(\frac{z_1}{z_0}, \frac{z_2}{z_0} \right), \quad \phi_1 = \left(\frac{z_0}{z_1}, \frac{z_2}{z_1} \right), \quad \phi_2 = \left(\frac{z_0}{z_2}, \frac{z_1}{z_2} \right)$$

is an atlas, and determine the domains of transitions such as

$$\psi(z_1, z_2) = \left(\frac{1}{z_1}, \frac{z_2}{z_1} \right).$$

(3) Show

$$\psi^* \omega_{FS} = \omega_{FS}.$$

(a) Show

$$\psi^*(-dd^{\mathbb{C}}f) = -dd^{\mathbb{C}}(\psi^*f).$$

(b) Show

$$f \circ \psi = f - \log |z_1|^2.$$

(c) Show

$$-dd^{\mathbb{C}}g = 2i\partial\bar{\partial}g, \quad \partial\bar{\partial}g = \sum_{j,k} \frac{\partial^2 g}{\partial z_j \partial \bar{z}_k} dz_j \wedge d\bar{z}_k.$$

(d) Show

$$\partial\bar{\partial}(-\log z_1 - \log \bar{z}_1) = 0.$$

Exercise 3.6 (1.11). For

$$u = (f_1 + ig_1, f_2 + ig_2): \mathbb{C} \rightarrow \mathbb{C}^2,$$

write

$$du \circ i = i \circ du$$

in real components and identify the classical Cauchy–Riemann equations.

Exercise 3.7 (1.19). Let

$$X = \mathbb{CP}^2 \# n \overline{\mathbb{CP}}^2, \quad H_2(X; \mathbb{Z}) = \langle h, e_1, \dots, e_n \rangle.$$

(1) For $A = a_0 h - \sum_i a_i e_i$, determine

$$\langle c_1, h \rangle, \quad \langle c_1, e_i \rangle, \quad \langle c_1, A \rangle.$$

(2) Determine the genus of a smooth symplectic representative of

$$3h - 2e_1 - e_2 - e_3.$$

(3) Show that $3h - 5e_1$ has no smoothly embedded symplectic representative.

(4) Show that $-3e_1 - e_2 - e_3$ has no smoothly embedded symplectic representative.

Exercise 3.8 (1.24). Show that at most one degree-2 J -holomorphic curve in $\mathbb{CP}^2$ passes through five fixed points.

Exercise 3.9 (2.5). Let

$$C = \{x^3 - y^2 = 0\} \subseteq \mathbb{C}^2, \quad B = \{((x, y), [u : v]) \in \mathbb{C}^2 \times \mathbb{CP}^1 \mid xv - yu = 0\}.$$

- (1) In the chart $u = 1, y = xv$, compute the proper and total transforms, the exceptional divisor, its multiplicity, and all intersections with the total transform.
- (2) Repeat in the chart $v = 1$.
- (3) Blow up the origin of the $u = 1$ chart and compute both charts, both transforms, and every intersection with the exceptional divisors.
- (4) Continue blowing up until the total transform has only transverse double intersections.
- (5) Repeat for

$$\{x^p = y^{p-1}\}.$$

- (6) Repeat for

$$\{x^3 = y^5\}.$$

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